Ujjwal Keshri

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Summary

Final-year Electrical Engineering student with a background in power system simulation and exposure to industrial PLC-based motor control, SCADA and VFD systems through an internship at Indian Oil Corporation. Developed fault and cyber-attack detection algorithms in MATLAB/Simulink as an academic project.

Work Experience

Industrial Trainee — Internship June 2025 - July 2025

Indian Oil Corporation Limited — SERPL, Bhubaneswar Pump Station (Jatni)

- Studied electrical distribution systems, transformers, protection relays, and busbar/feeder configurations at a critical petroleum pipeline pumping station.
- Worked hands-on with PLC-based motor control panels, SCADA I/O and HMI configuration, and VFD start/stop sequencing for pipeline pump units.
- Reviewed protection schemes, earthing systems, fire protection (HVWS), and safety compliance procedures under CEA regulations.

Projects

Power Grid Cyber-Attack Detection via V-I Correlation Link to Project

- Modelled a 5-bus AC transmission network in MATLAB/Simulink with normalised per-unit baselines to calibrate detection thresholds.
- Implemented detect_fdi() classifying system state as Normal / Cyber-Attack / Physical Fault at 50 µs sample rate using six FDI attack modes and a weighted event-scoring counter.
- Built classify_physical_fault() sub-classifier with four fault codes using RMS buffers and phase-unbalance metrics; detects faults within one cycle (20 ms).

Education

2023 - present	BTech-Electrical Engineering	(CGPA: 8.47/10)
	Odisha University Of Technology and Research,Bhubaneswar	
2022	Class 12th CBSE	(88.60%)
2021	Class 10th CBSE	(89.80%)

Skills

Technical Skills

Design & Simulation Altium Designer, AutoCAD Electrical, MATLAB & Simulink

Soft Skills

Problem-Solving, Analytical Thinking, Detail-Oriented, Self-Motivated